

Architecture Design Report

This document presents the architectural design of the system, including its context, logical structure, runtime behavior, and deployment considerations.

1. System Requirements

System Name: ResuFit: Precision CV Matching System

Architecture Style: Event-Driven/ Messaging

Functional Requirements

- System Functions
- Detailed Functional Specification
- User Login
- Lecture Upload
- File to Text Conversion
- Text Extraction Module
- Question Generator
- Question Bank
- Difficulty Level Selection

Non-Functional Requirements

- Security

Description: The system shall ensure secure access to user data and system features, and shall comply with relevant security standards.

- Usability

Description: The system shall provide a user-friendly interface that improves usability and facilitates the creation and management of content.

- Performance

Description: The system shall optimize performance to ensure efficient and reliable operation.

- Compatibility

Description: The system shall ensure compatibility with various file formats and systems.

- Scalability

Description: The system shall be scalable to accommodate increasing user demand and data volume.

- Reliability

Description: The system shall ensure reliable operation and minimize downtime.

- Maintainability

Description: The system shall be designed for easy maintenance and updates.

2. Formal Architecture Specification (ACME ADL)

The following section provides a formal architectural specification using ACME ADL, capturing components, con-

```
System ResuFit:PrecisionCVMatchingSystem {
  Property view : String = "runtime";
  Property architectureStyle : String = "Event-Driven/ Messaging";
  Property source : String = "AI";
  Component AuthenticationService = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Handles user login and authentication";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component LectureProcessor = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Processes uploaded lectures for question generation";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component FileConverter = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Converts uploaded files to text format";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component TextExtractionModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Extracts relevant text from uploaded files for question gene";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component QuestionGenerator = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Generates questions based on extracted text and user input";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component QuestionBankManager = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Stores and manages generated questions for future use";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
}
```

```

}
Component DifficultyLevelService = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Allows users to select the difficulty level of generated que
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component UserInterface = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Provides a user-friendly interface for users to interact wit
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component QuestionBankDatabase = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Stores and manages generated questions in a database";
    Property state : String = "stateful";
    Property scaling : String = "vertical";
    Property consistency : String = "eventual";
    Property replication : String = "multi-node";
}
Attachment AuthenticationService.out to UserInterface.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment UserInterface.out to QuestionGenerator.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment UserInterface.out to DifficultyLevelService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment QuestionGenerator.out to QuestionBankManager.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment QuestionGenerator.out to QuestionBankDatabase.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment FileConverter.out to TextExtractionModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment TextExtractionModule.out to QuestionGenerator.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}

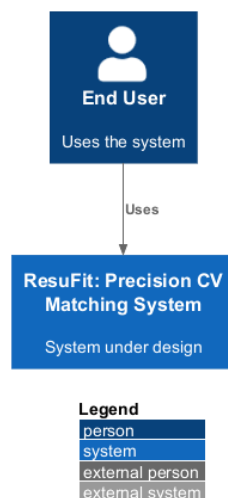
```

```

}
Attachment LectureProcessor.out to TextExtractionModule.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = true;
  Property backpressure : String = "bounded";
}
Attachment DifficultyLevelService.out to QuestionGenerator.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = true;
  Property backpressure : String = "bounded";
}
Attachment QuestionBankManager.out to QuestionBankDatabase.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = true;
  Property backpressure : String = "bounded";
}
Property resilience_pattern : String = "Retry, CircuitBreaker";
Property delivery : String = "at-least-once";
Property ordering : Boolean = false;
Property authentication : String = "service-to-service";
Property authorization : String = "RBAC";
}

```

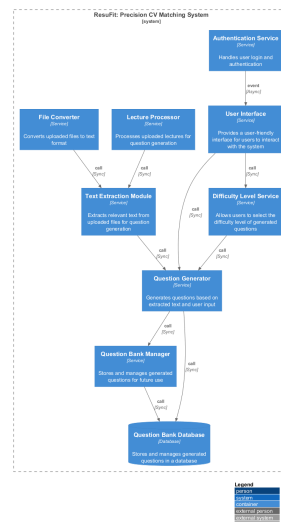
3. Context View



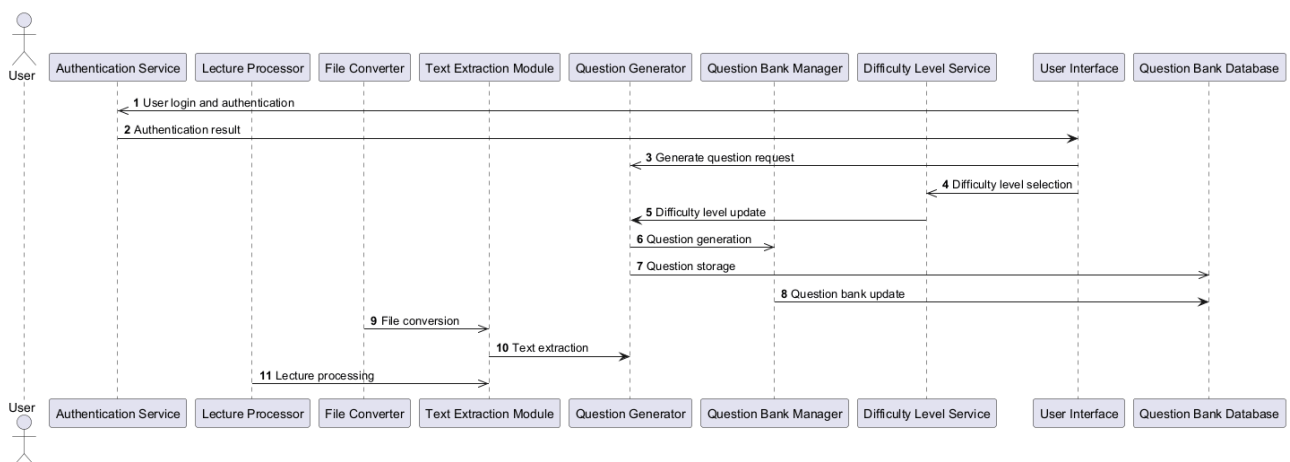
3.1 Data Context View (Level 0 DFD)



4. Logical View (C4 Container Diagram)



5. Process View (Runtime Interaction)



6. Physical View (Deployment Diagram)

